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SmartKitchen.AI

AI-powered recipe suggestion from ingredient images

Kezia Kunju John

Gifty Augustine

Divya Sabu

Introduction & Vision

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Problem Statement:

The "What's for dinner?" dilemma. Users have ingredients but lacks inspiration or knows the exact recipe names.

Solution

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“SmartKitchen.AI”

Visual first tool bridging gap between raw ingredients and executable recipes

Technology Stack -01

- **Backend Architecture**:
  - FastAPI: High-performance Python web framework.
  - Uvicorn: ASGI server for asynchronous execution.
- **Frontend Design**:
  - Single Page Application (SPA): Using modern vanilla JavaScript and CSS.
  - Design Language: Glassmorphism, animations (Inter font, FontAwesome icons), and responsive layouts.

Technology Stack -02

- **AI Core**:
  - **QwenVL-2:** A powerful Vision-Language Model tuned for ingredient recognition.
  - **Transformers & Torch:** Underlying libraries for model inference.

Data Engineering & Preprocessing -01

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Recipe Dataset

Utilizing the Recipe1M+ dataset (1 million recipes, 13 million images).

Visual Data

- **Captured own images** of fridge or kitchen ingredients
- 35 test photos with mixed raw ingredients to **test model robustness** on realistic, unstructured scenes
- Random images from the internet.

Data Engineering & Preprocessing -02

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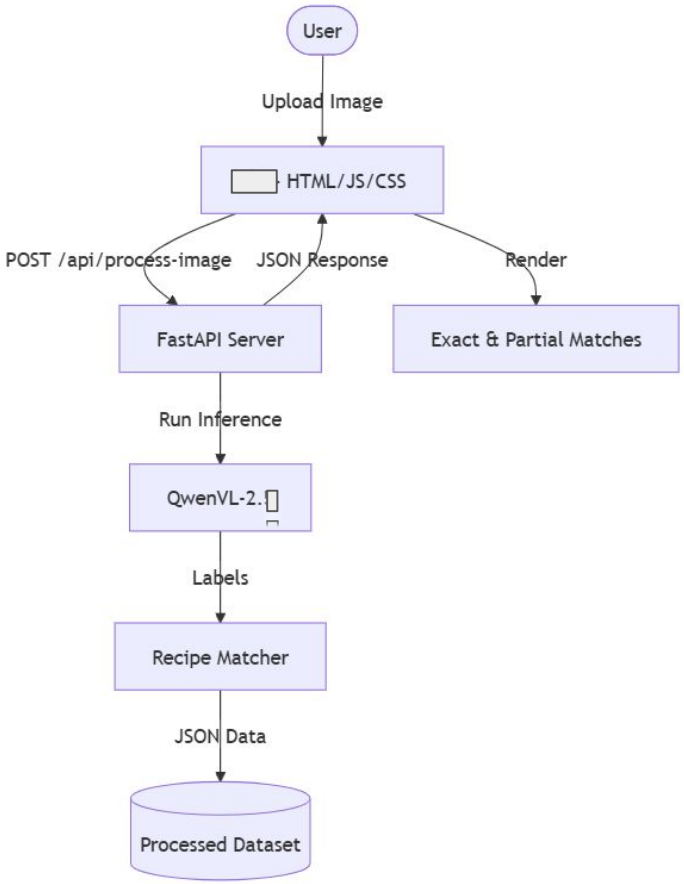
Processing Pipeline (clean_dataset.py):

- Filtering "non-dish" items (e.g., powders, extracts).
- Ingredient normalization (unit stripping, descriptor removal).
- Heuristic-based cleaning to ensure high-quality matching.

Inventory Analytics (generate_glossary.py):

- Frequency analysis of ingredients across 100k+ recipes.
- Automatic singularization and merging of plural forms.

Architecture - 01



Architecture - 02

VLM Analysis Flow:

Image upload

Encoding

QwenVL Inference

Tag extraction.

The Matching Algorithm

- **Exact Matches**: Recipes where the user has 100% of required non-staple ingredients.

- **Partial Matches**: Ranking recipes based on "Coverage Score" (>20%) and "Missing Count".

Workflow - 01

1. **Visual Capture**: The user captures or uploads a photo of their kitchen counter or fridge.
2. **Vision-Language Inference**: The QwenVL-2 model parses the scene, identifying specific ingredients (e.g., "bell pepper," "chicken breast").
3. **Semantic Normalization**: Our cleaner script strips units and pluralizes/singularizes terms to ensure they match our database vocabulary (e.g., "carrots" -> "carrot").

Workflow - 02

4. Heuristic Filter Matching: We filter millions of recipes using a multi-tier logic.

5. Interactive Results: The frontend renders recipes with "Match Scores" and highlights "Missing Ingredients" to help the user decide.

Project Milestones & Work Accomplished - 01

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1. **Dataset Acquisition & Research**

   Identified and integrated the **Recipe1M+** dataset as our primary source.

2. **Intelligent Pre-processing**

- **Cleaning**: Developed clean_dataset.py to strip noisy descriptors (e.g., 'chopped', 'fresh') and non-food items.
- **Glossary Engineering**: Built an automated ingredient registry with plural-merging logic to ensure semantic consistency.

Project Milestones & Work Accomplished - 02

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3. VLM Evaluation & Selection

- **Benchmarking**: Evaluated multiple Vision-Language Models for ingredient recognition accuracy.
- **Optimization**: Selected **QwenVL-2** for its superior performance in identifying raw food items across diverse kitchen settings.

Project Milestones & Work Accomplished - 03

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4. SmartKitchen.AI Implementation

- **Backend:** Created a fast backend using FastAPI for handling VLM inference.
- **Frontend:** Built a simple, modern UI with a glassmorphism design for easy interaction.
- Logic: Implemented the weighted "Exact vs. Partial" matching algorithm.

Evaluation Strategy

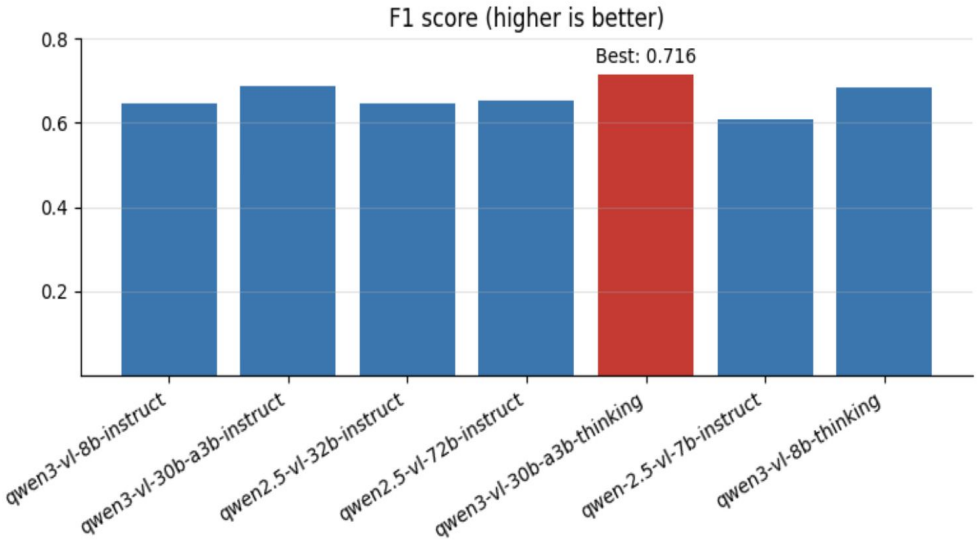
Since this is a practical AI project, evaluation is partly qualitative:

Evaluation Aspect	Metric / Method
Ingredient detection	Precision, recall, F1
Retrieval Testing	Overlap between detected and recipe ingredients

Results : Qwen Model Comparison

Model	Precision	Recall	F1 Score
qwen/qwen3-vl-8b-instruct	0.630	0.672	0.647
qwen/qwen3-vl-30b-a3b-instruct	0.662	0.726	0.687
qwen/qwen2.5-vl-32b-instruct	0.606	0.702	0.645
qwen/qwen2.5-vl-72b-instruct	0.636	0.691	0.654
qwen/qwen3-vl-30b-a3b-thinking	0.695	0.749	0.716
qwen/qwen-2-vl-2b-instruct	0.568	0.534	0.540
qwen/qwen3-vl-8b-thinking	0.659	0.721	0.686

Final model choice for deployment



Qwen2

We built the website using Qwen 2 (VL) for ingredient understanding + generation.

Even though a Qwen3-VL model had the best F1 score during the test , we choose a Qwen2VL for the final system based on deployment practicality (latency, stability and availability).

Challenges

1. Scaling Challenges: Transitioning from JSON-based matching to SQLite FTS5 for million-record lookups.
2. DevOps/Deployment: Handling port conflicts (8000 vs 8001 vs 8002) and path robustness on Windows systems.
3. Poor-quality or cluttered images from user
4. Images may include personal or identifiable items
5. Ingredient names vary by language or region

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Demo!!

References

- Qwen2-VL Model: https://huggingface.co/Qwen/Qwen2-VL-2B-Instruct
- OpenRouter API: https://openrouter.ai
- Datasets:Recipe1M+, Kaggle

THANK YOU!